JM-Orgue

User guide for version 0.9.56 · November 2026

JM-Orgue is a virtual pipe organ for Windows, macOS and Linux. It plays Hauptwerk and GrandOrgue sample sets directly, including large surround sets and Hauptwerk Custom Organ definitions, and you play it with a MIDI keyboard, an organ console or the mouse. It can write down what you play as you go, in staff notation or in klavar, and print it as a PDF. This guide walks through the program; the screenshots come from version 0.9.16 and later versions, the text describes version 0.9.56.

© Martijn van der Kolk. The screenshots were made with demo organs and own test sets; network addresses, access tokens and user folders have been made unreadable.

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1. Getting started

Installing

Download the installer of the latest release from the project's release page (for Windows `JM-Orgue_<version>_x64-setup.exe`) and run it. The installation is per user and needs no administrator rights. There is also an MSI package for Windows, a DMG for macOS and an AppImage or deb for Linux.

The program itself is free to use. The sample sets are not included: you get those from their makers. The start page of the library lists a few makers who offer sets free of charge.

Updating

JM-Orgue checks by itself whether a new version is available. If there is one, a notice appears at the top. Choose **Update now**: the program downloads the update, checks its signature, installs it and restarts. If you only want to download first, choose **Download**. You can also check by hand via App Settings → Update and feedback → **Check for updates**.

The three main screens

At the top are three tabs: **Organ** (playing), **Organ Settings** (everything that belongs to the loaded organ) and **App Settings** (everything that applies to the whole program). `F1`, `F2` and `F3` switch between them directly. The cross next to the organ name closes the organ and takes you back to the library.

2. The library

After starting you see the organ library: one card per organ with photo, name, builder, place and number of stops. Click a card to load the organ. The grid shows eight cards side by side and three rows in view; if the library holds more organs, you scroll with the two round arrow buttons on the right (**Scroll up**, **Scroll down**) or with the scroll wheel. The buttons are large enough for a touch screen and disappear as soon as everything fits in view.

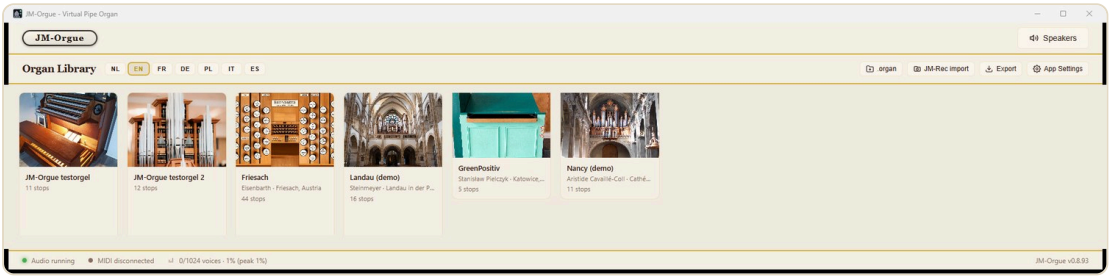


Figure 1. The library with demo organs. At the top right are the buttons to add organs and to go to the settings; at the top left the language choice (NL, EN, FR, DE, PL, IT, ES).

Finding sample sets

Below the cards is the block **More sample sets** with buttons to makers who offer sets free of charge: Piotr Grabowski, Sonus Paradisi, Lars Palo and Augustine's Virtual Organs. Each button opens that maker's download page in your browser; JM-Orgue reads their Hauptwerk and GrandOrgue files directly, without conversion. Next to them is the **Manual** button, which opens this guide as a PDF in the language of the program. With an internet connection the library also shows, under **Available to download**, a card for each set the project offers itself: click it, choose a folder, and the set is downloaded, unpacked and added to the library.

Adding an organ

- **.organ**: choose an organ definition (GrandOrgue `.organ` or a Hauptwerk definition). You choose the definition file, not the loose folder of samples: only the definition knows the stops, couplers, tremulants and any extra recording positions.
- **.orgue** and **.rar**: choose a GrandOrgue package (`.orgue`) or a RAR archive (`.rar`), including a multi-part archive: choose one of the parts and keep all parts in the same folder. JM-Orgue unpacks it once into its own `pakketten` folder, shows the progress in the loading bar and loads the organ; opening it a second time costs nothing. If a part is missing, the archive is password-protected or damaged, a clear message follows. The unpacked folder takes disk space: you can delete the `pakketten` folder in the program's data folder to free it.
- **JM-Rec import**: import a sample set made with JM-Rec.

- **Export**: turns a folder of samples into an organ definition (`.organ`), which you can then load with the **.organ** button.

Card buttons: photo, search, remove

If you point at a card, three buttons appear at the top right of the photo: **search a photo online** (globe), **choose an image yourself** (camera) and **remove from library** (cross). Removing only takes the card out of the library; the files on your disk stay where they are. If you hold a card pressed for 3 seconds, the program searches the photo again.

Searching online shows a few freely usable photos from Wikimedia Commons. At the top of the window are the fields **Church**, **Place** and **Builder**, filled in beforehand from the organ definition and the information supplied with the set; correct them if they are wrong and choose **Search**. Below them a free search field remains for a term of your own. Click a photo to adopt it; the source and licence appear under the card.

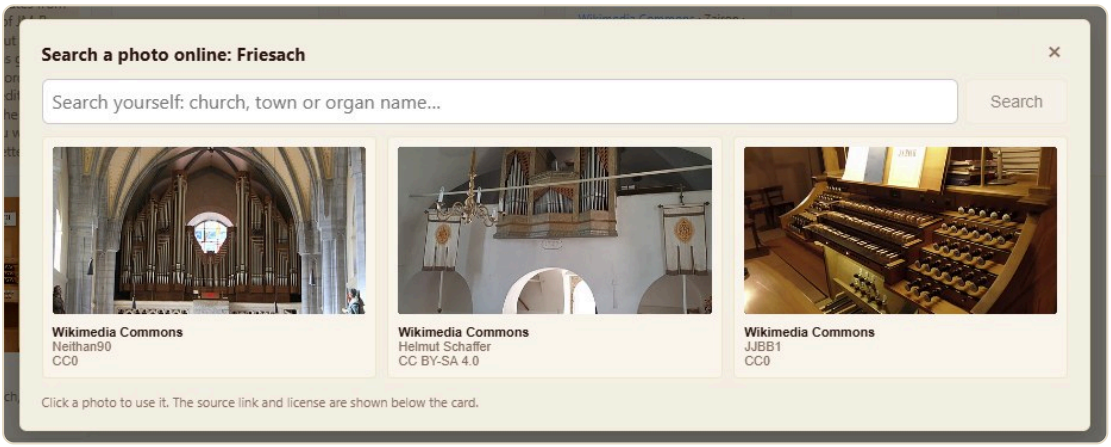


Figure 2. The “Search a photo online” window with the field for your own search term and the photos found, with their source and licence.

Downloadable sets

Sets you can fetch with the download button in the program appear as a tile with a dotted border in the same grid as your organs. A set you have already installed is no longer offered.

Hauptwerk Custom Organ (CODM) and the import report

A Custom Organ definition (`.CustomOrgan_Hauptwerk_xml`) only describes the structure of the organ: keyboards, stops, couplers and tremulants. The samples live in separate **sample packages** in the `OrganInstallationPackages` folder, which must sit next to or above the definition. JM-Orgue checks this before loading.

- If a sample package is missing, the **Organ is incomplete** screen shows the missing packages (by name) and how many stops would sound. The organ you have now stays active until you choose.
- **Choose package folder…** lets you point to the folder that contains `OrganInstallationPackages`; the choice is remembered per organ. No files are copied or converted.
- **Load partially** loads only the stops whose samples are present; a bar at the top reminds you and opens the **Import report**. If nothing is playable you can only cancel.
- The **Import report** lists what was expected and what was loaded, which stops were skipped and why, and where JM-Orgue uses an approximation (for example: the tremulant sounds as a simple

modulation when the set's waveforms cannot be read).

- JM-Orgue cannot read samples in Hauptwerk's protected `.hbw` format; a set that depends on them is reported as not playable.
- A stop with only one pipe (cymbelstern, nightingale, a single bell stroke) does not go on the keyboard but becomes a clickable button in the **Effect registers** division, just as with regular Hauptwerk sets. Stops with more pipes stay on the keyboard.
- Shortcut pistons from the definition get no buttons of their own: they operate a stop, coupler or tremulant that already has a button in its division. The report lists, per piston, what it points to.
- Couplers behave as the Hauptwerk guide prescribes: a coupler connects a keyboard to the wind chests of its target and does not chain on to a further coupler. GrandOrgue sets do keep chaining.
- If the definition is written in the "abbreviated" form (fields are then only called `a`, `b`, `c`), the report states whether the meaning of those fields could be derived safely.

A CODM organ with all its samples loads without any question. If the definition states per stop that the upper harmonics should sound stronger or softer, JM-Orgue applies that while loading. Whether it sounds exactly as in Hauptwerk has not been verified: that needs a complete original set and a reference recording.

The import report also shows, as information only, what the set itself provides: the swell box values, any crescendo, the recorded tremulant waveforms and the wind system (the bellows with their natural frequency and pressure, and which divisions are attached to them). The tremulant can follow those waveforms and the wind model the wind system (see chapter 7, Behaviour of the set).

Sample sets with WavPack or FLAC

Many GrandOrgue sets are packed with WavPack; the files are then often simply called `.wav`. JM-Orgue reads them directly, including loop points and releases, and loads them quickly because only the beginning of each recording is unpacked. Only if a file really cannot be read (for example a FLAC file or a multichannel WavPack file) does a bar at the top report how many files were skipped.

3. Loading and playing an organ

Loading

When loading, JM-Orgue reads the beginning of every pipe recording into memory; the rest is loaded while you play. A large surround set can therefore take a moment; the progress bar shows how far it is. Below the indicator is a **Cancel** button: press it while the samples are still being read and the program returns to the organ that was already sounding, without changing anything. Handy if you clicked the wrong card by accident. Once the audio engine is already switching over, the button does nothing any more.

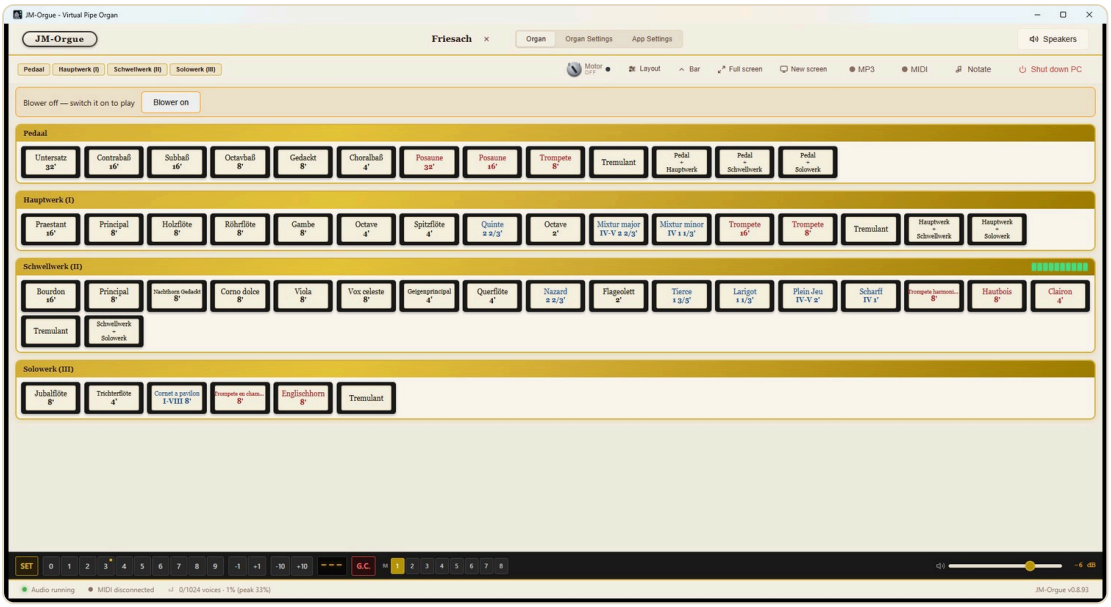


Figure 3. The loading screen with the progress of reading the samples.

The blower

After loading, the **organ blower is on by default** as soon as the set has a blower recording, and you hear a soft wind right away. To practise in silence without that wind sound, switch the **Blower** button in the toolbar off; the yellow bar only appears when the blower is truly off. Which background sounds are on by default and how loud, you adjust from the toolbar with the small mixer next to the background-sounds button.

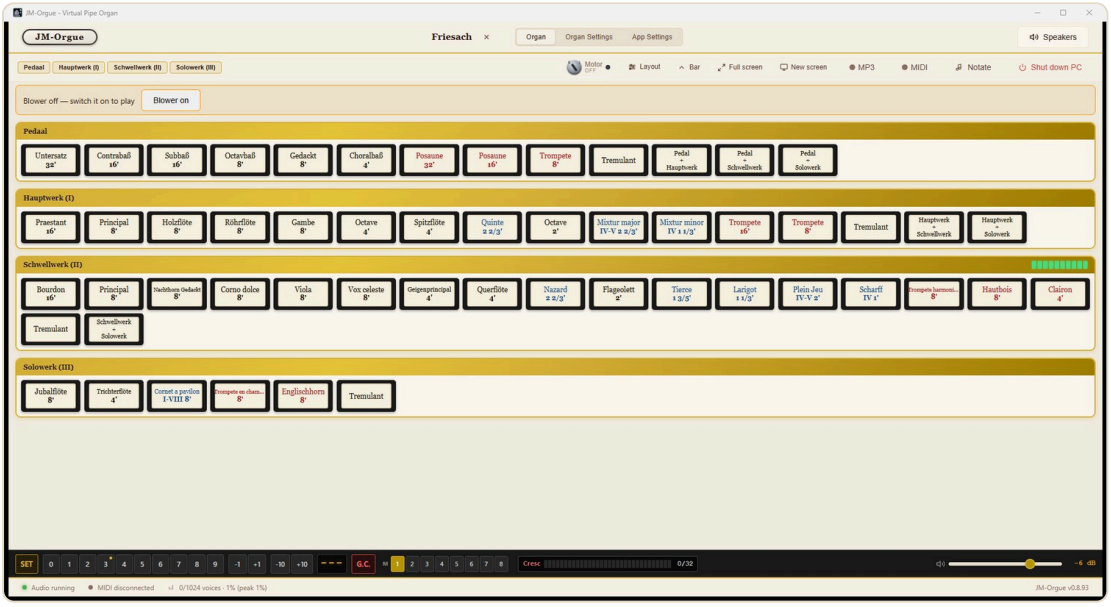


Figure 4. The organ screen right after loading: blower off, all stops pushed in.

Stops, couplers and tremulants

Every division has its own box with its stops. Click a stop knob to draw it; drawn stops turn yellow. The colour of the name shows the family: reeds, for instance, are red, mixtures and quints blue. If the automatic classification is wrong for a single stop, you choose the colour group yourself under Stop order (chapter 7). Each box also holds the **tremulant** and the **couplers** (for example “Hauptwerk + Schwellwerk”); the couplers use the same font size as the stops. A green bar on the right in the header

of a division shows the position of the swell box. The divisions appear in the order of the organ's keyboards (I, II, III), not in the order of the file. If the set has effect stops (cymbelstern, nightingale and the like), they appear as clickable buttons in a division of their own, **Effect registers**.

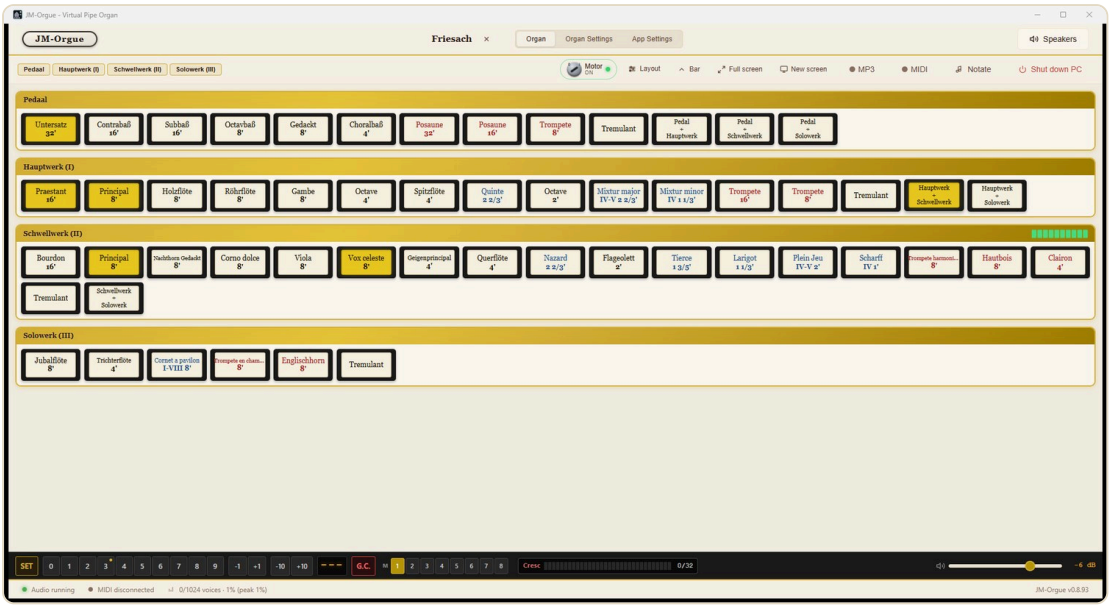


Figure 5. Blower on, a few stops drawn and the coupler Hauptwerk + Schwellwerk on. The setter bar at the bottom.

The setter bar

At the bottom is the setter bar, with the controls of an organ console:

Item	How it works
SET and **0–9**	Press SET and then a digit to store the current registration; press just the digit to recall it. A small dot at a digit means a registration is stored there.
−1 / +1, **−10 / +10**	Step to the previous or next registration, or ten further.
G.C.	General Cancel: all stops off.
M 1–8	Memory levels: eight separate banks of registrations.
Copy button	**Copy combination to…**: recall a combination, press the copy button, choose the destination with −10 / +10 and a digit, and the combination is copied there, also across a block boundary (for example from 9 to 10). The registration that is sounding now does not change.
Cresc	The general crescendo (see chapter 7). The bar shows the current step.
Volume slider	Master volume. Double-clicking resets it to the default; with the scroll wheel you change it in steps of 1 dB.

Every button of the setter bar can be linked to a button on your own console: right-click (or hold for 3 seconds) and then press the button on your console.

Learn the MIDI channel of a manual

You do not have to visit Organ Settings first: right-click the name of a division above the stops — or hold it for 2 seconds on a touch screen — and the program asks for the lowest and highest key of that manual. The channel is set without leaving the console.

4. The toolbar

Above the stops is the toolbar. On the left are the divisions (Pedal, Hauptwerk, …) as buttons: with them you show or hide a division. On the right are the functions below.

Button	What it does
Blower	Organ blower on or off.
Layout	Opens the layout menu (see below).
Bar	Shows or hides the main bar with the tabs and the profile button.
Full screen	Console mode: only the stops and the setter bar (`F11`).
New screen	Opens an extra window with the same stops, for example for a second monitor or a second keyboard.
MP3	Records what you hear. The recording goes to Documents\JM-Orgue-opnames. Click the red counter to stop; the program then tells you where the file is.
MIDI	Records what you play (notes only) as a .mid file. Click the blue counter to stop and choose where to save it. The arrow next to the button opens a menu: **Open MIDI file…** picks a .mid file and gets it ready, and **Start recording** does the same as the button itself. Once a file is open (or has just been recorded), a **Play** button appears; it becomes **Stop** while the file plays. Draw the stops yourself: a MIDI file contains notes only. More options (pause, position, speed) are under Routing, in the block Play MIDI.
Notate	Opens the notation window: notes appear while you play, in staff notation or klavar, with staves, overdub and editing, and can be printed straight to PDF (see below).
Shut down PC	Closes the software *and the computer*, after a confirmation; the last state is saved first. Right-click or hold for 3 seconds links the button to a button on your console.

The layout menu

With **Layout** you adapt the display to your screen: **Fit** measures the button size so that all stops fit in view, the minus/plus buttons change the width and the height of the buttons, and you further choose divisions below or beside each other (vertical), round instead of rectangular buttons and the place of the couplers (in the division or in a bar of their own). The choice is remembered per organ.

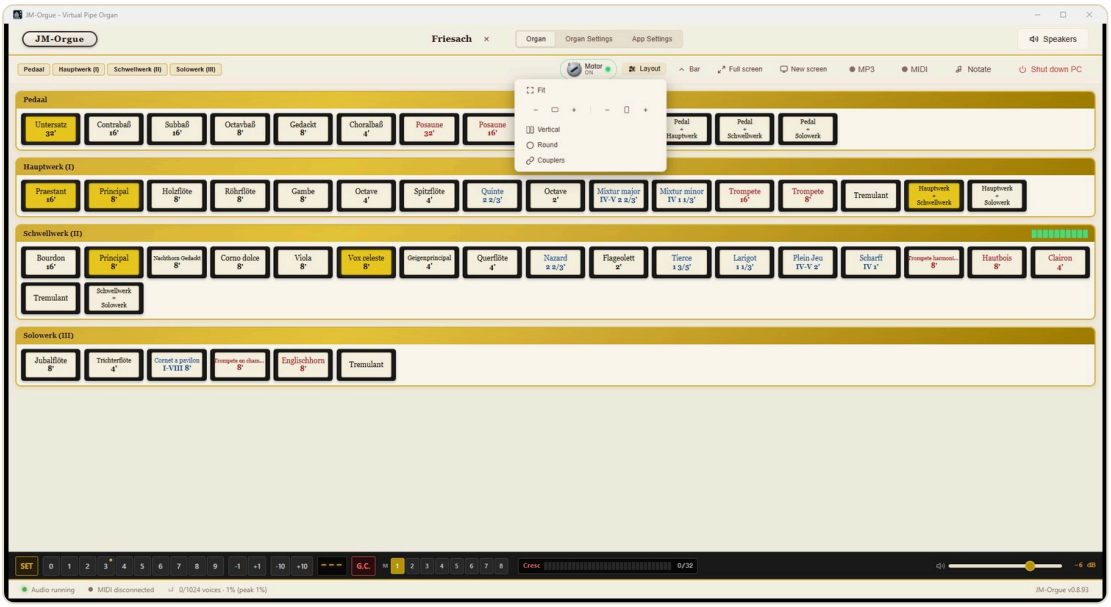


Figure 6. The Layout menu in the toolbar.

Full screen (console mode)

In this mode the title bar, the Windows taskbar and the main bar disappear: you see only the stops and the setter bar. At the top is a narrow strip with four buttons: **Bars** (bring the main bar back temporarily), **Blower** (on or off), **Exit full screen** and **Shut down PC** (close the software and the computer). `F11` switches the mode on and off; `Esc` turns it off.



Figure 7. Console mode: stops, setter bar and at the top the strip with the four buttons.

A second screen

New screen opens a window of its own that you can drag to another monitor, or leave small beside the main window. It has its own layout settings: if the buttons are too small, choose **Layout** and set the button width and height, or choose divisions below or beside each other.

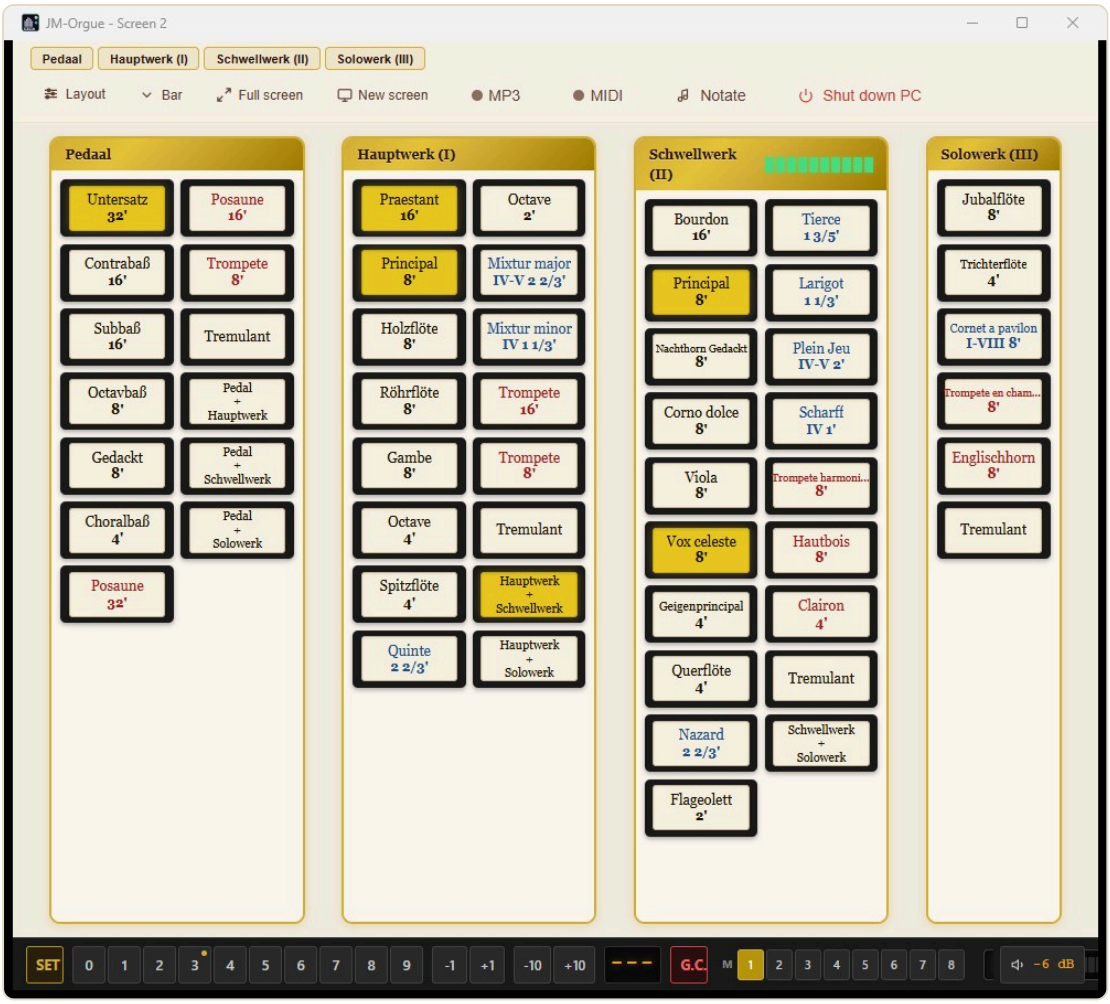


Figure 8. An extra screen with readable stops, here in a window of 1000 × 900 pixels.

The notation window: staff notation, klavar and PDF

Notate opens a window of its own in which your playing is written down; no second program is needed to compose or to work out a piece. Choose **New…** for a new piece (title, composer, time signature, key, tempo, number of bars and staves) and press **Record**: recording starts at the first note you play. A piece has several **staves**, for example right hand, left hand and pedal; under **Staves & voices** each staff has its own **Channel** with a **Learn** button (play one key on the keyboard that should go to that staff). If you play right hand, left hand and pedal on three MIDI channels, they land on three staves at the same time. A new take (overdub) is added to what is already there.

- **View**: **Staff notation** shows ordinary staves; **Klavar** shows klavarskribo, in which the lines are the black keys and time runs from top to bottom. Per staff you choose the hand (right, left, pedal, or R+L with a split point) and with **Range** whether the staves span the notes played or the whole keyboard.
- **Print / PDF** prints the score or saves it as a PDF, in the chosen view. Under **Export** you also find **Save as MIDI…**, **Save as MusicXML** (for other notation programs) and, for klavar, **Save as SVG…**.
- **Save** stores the piece as a JM-Orgue score (`.jmscore`), **Open…** brings it back; an existing MIDI file can be imported as an extra take. On the tabs **Input**, **Texts & marks** and **Playback** you enter notes step by step, add texts and marks, and play the piece through the organ.

Keyboard shortcuts in the notation window: `Ctrl` + `N` new piece, `Ctrl` + `O` open, `Ctrl` + `S` save, `Ctrl` + `C` copy.

5. Several listening positions

Some sample sets were recorded at several places in the church, for example close, in front, halfway and at the back. Such a set shows the **Perspective** buttons in the toolbar, one for each position. Click a position to switch it on or off; you can let more than one sound at the same time. To the right of them is the **listening position**: the slider between “at organ” and “in the back” blends smoothly between the recorded positions.

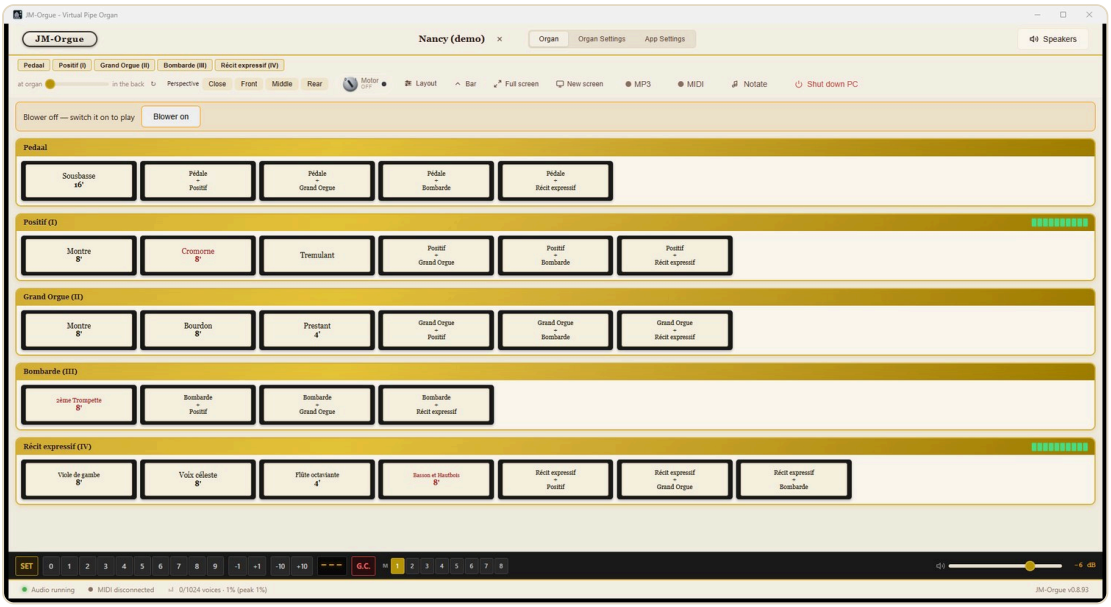


Figure 9. A set with four recording positions (Close, Front, Middle, Rear) and the listening-position slider.

> **TIP** Next to the Perspective buttons in the toolbar is a mixer button: click it for a small mixer with a slider per position (0–200 %). Which positions are loaded you choose under Organ Settings → “Perspectives and stacked ranks”; only the ticked positions go into memory, and that choice takes effect after the organ is reloaded. **Keep all positions in memory** is off by default and is only carried out when all positions together fit comfortably in memory. If an organ does not fit in memory, the program switches on only the nearest position after loading and reports that at the top.

6. The setup wizard for a new organ

If you load an organ for the first time, the setup wizard “Set up organ” starts automatically. In six steps it goes through everything you have to connect for your own console. Everything can be changed later, and you can start the wizard again via Organ Settings → Routing → **Wizard**. **Later** closes the wizard and asks again the next time you open the organ; **Skip** (step 1) skips it altogether.

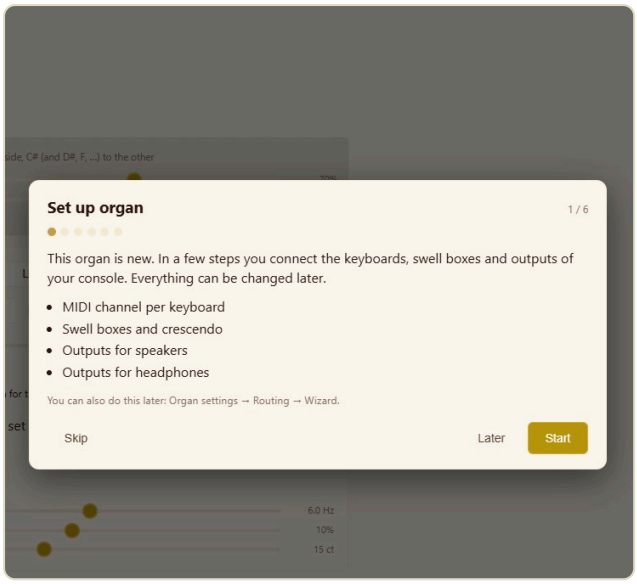


Figure 10. Step 1: what the wizard does.

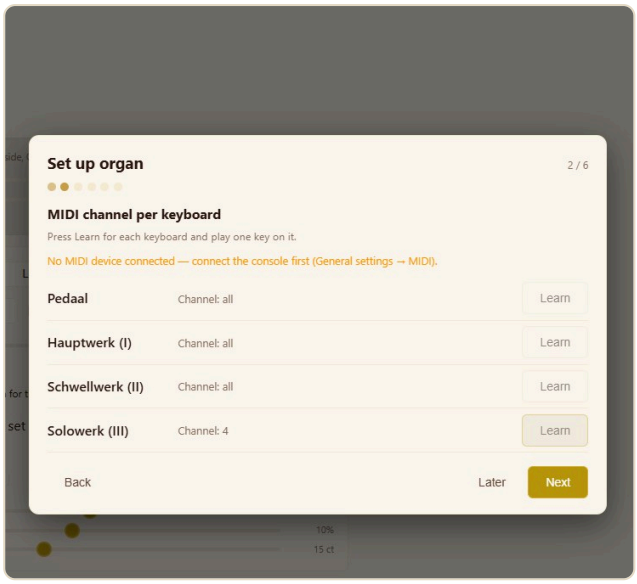


Figure 11. Step 2: the MIDI channel per keyboard.

1. **Start**: an overview of the steps.
2. **MIDI channel per keyboard**: for each keyboard choose **Learn** and play a key on that keyboard. The program remembers the channel. The buttons only work when a MIDI device is connected (App Settings → MIDI); otherwise a warning is shown.
3. **Swell boxes and crescendo**: choose **Learn** and move the pedal fully closed and fully open. The same goes for the general crescendo. If a set has swell boxes per rank (chapter 9), they appear here too.
4. **Outputs for speakers**: choose the driver and the device, save them as the “Speakers” profile and choose per keyboard which outputs of your audio interface it sounds on. Below that are the same choices for the blower sound, the ambient noise and action clicks, the effect stops and the tremulant noise, as far as the set has them.
5. **Outputs for headphones**: the same for the second profile, for example a headphone output. From then on you switch between the two profiles with the **Speakers** button at the top right.
6. **Done**: the settings have been saved with this organ.

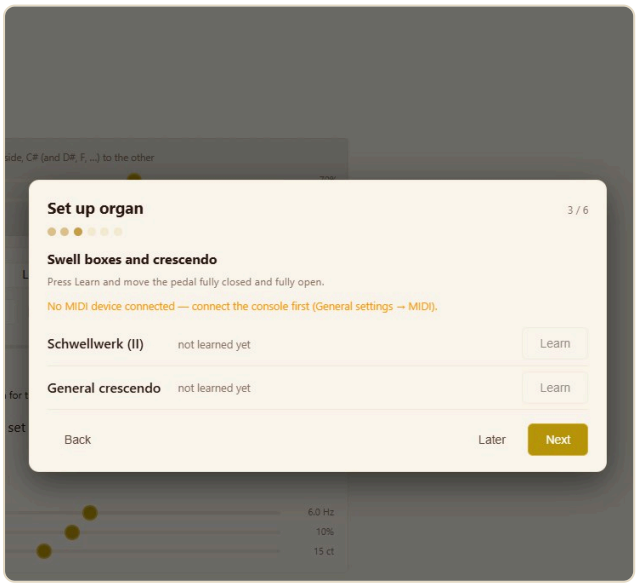


Figure 12. Step 3: learning swell boxes and crescendo.

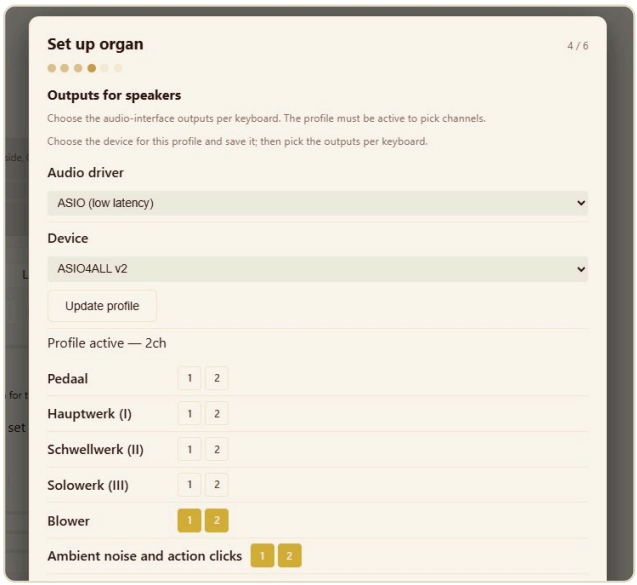


Figure 13. Step 4: driver, device, channels per keyboard and background sounds for the speakers.

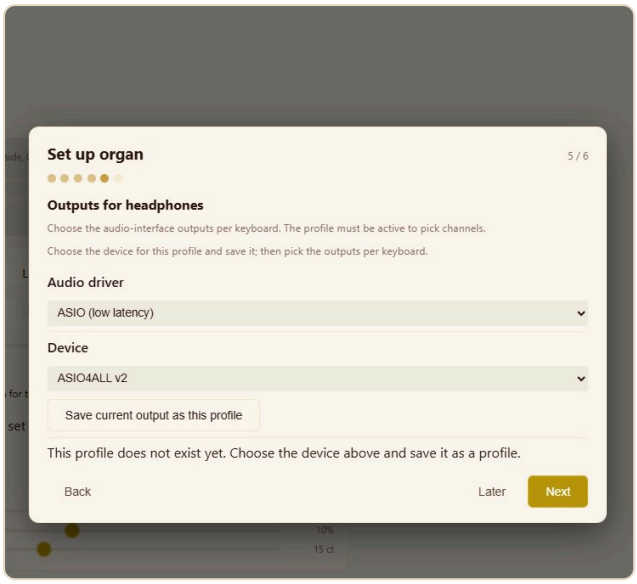


Figure 14. Step 5: the same for the headphones.

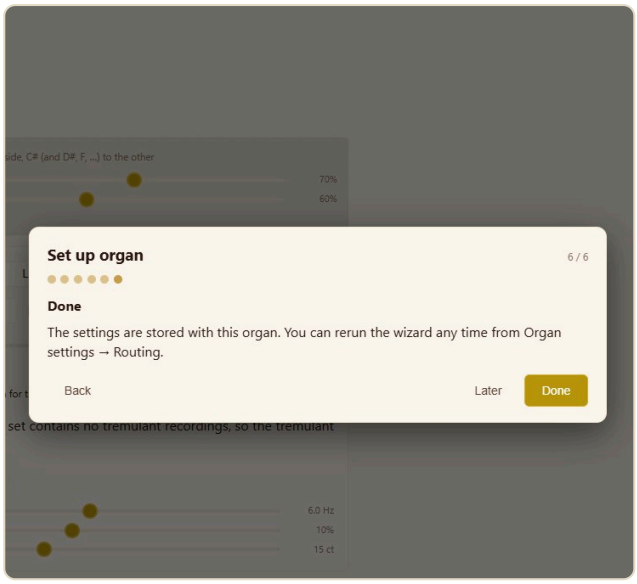


Figure 15. Step 6: done.

7. Organ Settings

Settings that belong to the loaded organ are stored per organ. At the top are the tabs **Sound**, **Loading**, **Crescendo**, **Stop order**, **Routing** and **All**; the last one puts all blocks below each other. Under **Loading** you choose the recording positions and stops that are loaded into memory.

Every organ definition gets its own settings file, even when several definitions are in the same folder. If the program finds an older file `.jm-settings.json` whose owner is unknown, it asks whether that file belongs to the chosen organ. The old file is kept. If you are not sure, choose Cancel; that choice is remembered for this organ on this PC, also after a restart. Under **Stop order → export/import settings** you can later choose **Import old organ settings…**. Existing settings are only replaced after an extra confirmation; the program first makes a backup of them.

Sound

The first block holds the master settings:

- **Volume**: the master volume of this organ.
- **Temperament**: "Original" plays the pipes as they were recorded. Other temperaments (equal and historical temperaments) retune the pipes based on the measured pitch of each recording. Below the choice you see how many pipes have a measured pitch. What the organ definition states takes precedence: a floating stop with a stated deviation (Unda maris, Voix céleste) sounds at the temperament plus that deviation, a recording whose pitch is given in the definition is retuned to it, and a set that states it does not play at the original organ pitch always plays at the target pitch. With **Custom temperament…** you set the deviation per note in cents yourself. **Load Scala file…** reads a .scl file (the worldwide format for microtonal and historical temperaments from the Huygens-Fokker Foundation); the loaded temperament appears in the list and is kept for later sessions.
- **Fine tuning**: everything together higher or lower, in cents.

Reverb controls the reverberation. If the set already has the church acoustics in its recordings, that is stated (with a button **Back to the recording (reverb off)** to switch the extra reverb off): extra reverb is then a matter of taste. You choose between **Algorithmic** (a calculated reverb with RT60, delay, damping and room size, also as a preset) and **Impulse Response** with an impulse response (IR) of your own. With **Mix** you determine how much reverb you hear.

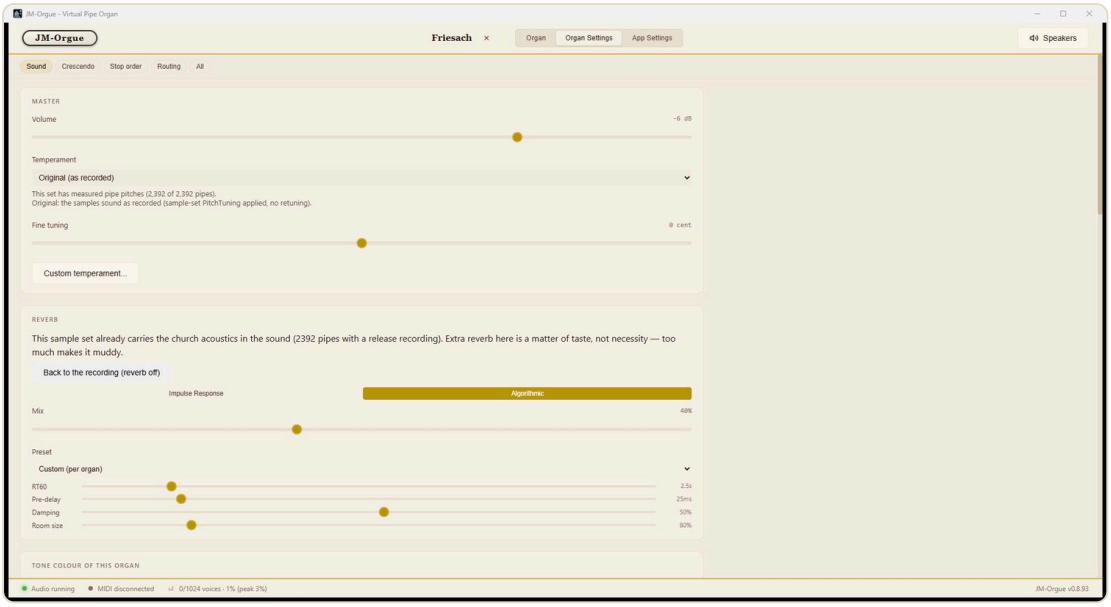


Figure 16. Sound, at the top: master (volume, temperament, fine tuning) and reverb.

Tone colour of this organ is a tone control of its own per organ; the correction for your headphones or speakers is separate, in the app settings. The **wind model** lets the organ breathe: a chord with many stops lowers the wind pressure, as in a real organ, and flutes and reeds react to it differently. You choose the number of wind groups (which divisions share the same bellows) and, per group, the bellows size, the damping, the maximum pressure drop and the character of the wind.

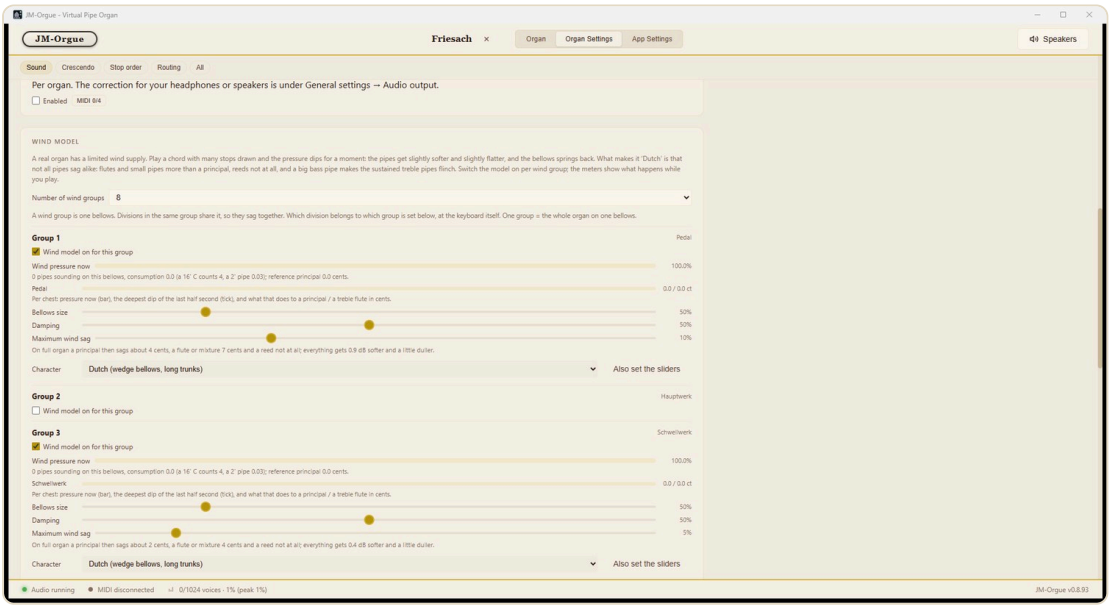


Figure 17. Sound, further down: tone colour of this organ and the wind model per wind group.

Background sounds shows the continuous sounds the set contains, such as the blower and room noise. You set the volume, switch the **Blower** on or off and can enable the **key action noise** (clicking

keyboard). The key action is off by default because it takes a lot of memory and requires reloading the organ.

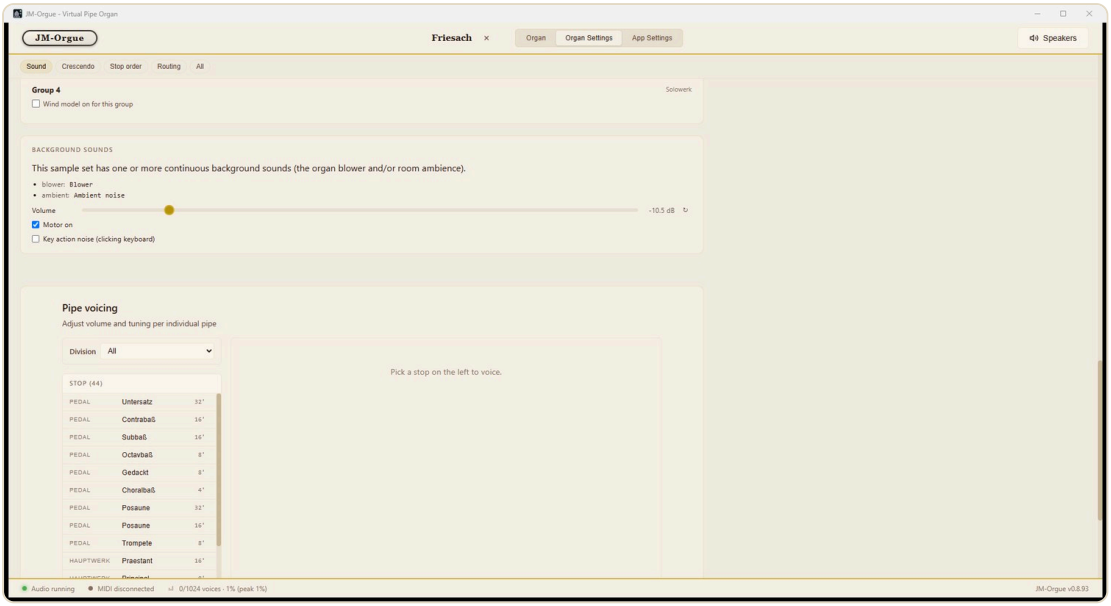


Figure 18. Sound: wind groups and the Background sounds block with the volume, the blower and the key action.

At the bottom is the **pipe voicing**: choose a stop on the left and adjust the volume and tuning of each individual pipe. To shift the whole stop in one go, set the sliders to the desired value and press **Apply to all pipes** — every pipe of that stop then receives the same volume and tuning adjustment.

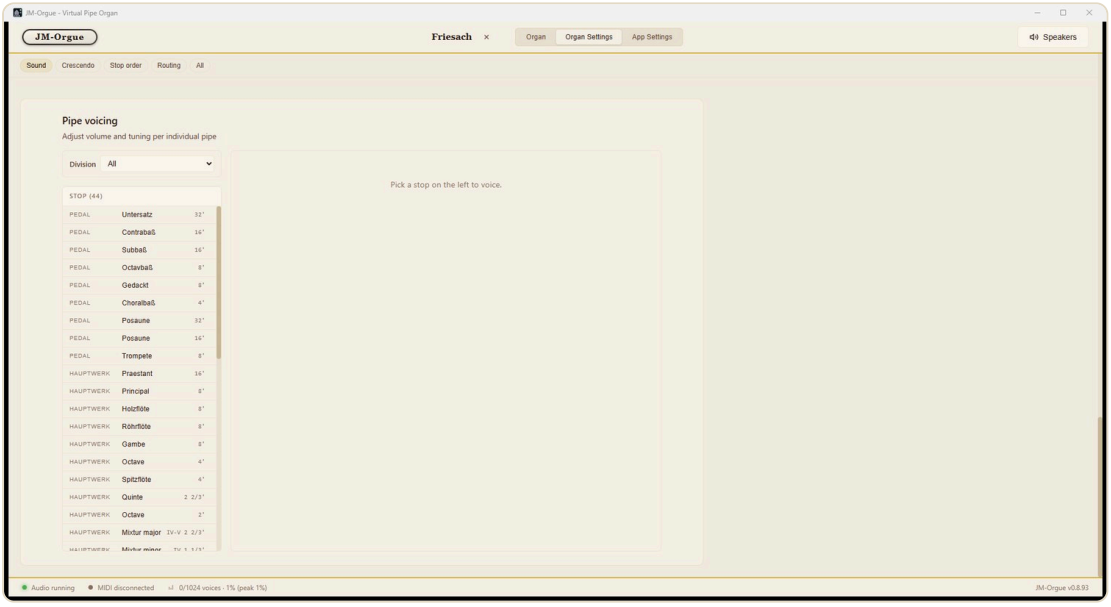


Figure 19. Sound, at the bottom: the pipe voicing. Choose a stop on the left to voice its pipes.

Behaviour of the set and not loading stops

Under **Sound** there is the block **Behaviour of the set (this organ)** with check boxes that decide whether JM-Orgue follows the data of the set itself: **Apply the set's own levels**, **Release crossfade of the set's maker**, **Couplers follow the set**, **Read all couplers from the set**, **Wind system of the set**, **Tremulant follows the set's waveform** and **Cut off silent tails** (with a threshold in dB). They apply per organ: an organ you already used stays as it was, a newly installed organ follows the set. **Use the set's swell box values** is always off by default; on means the swell box starts with the attenuation and filter

the set itself specifies, for divisions you have not set yourself. After changing a check box, reload the organ.

Read all couplers from the set also reads, for a Hauptwerk set, the couplers that point to another keyboard (including octave couplers and couplers within one keyboard) instead of guessing them. With **Wind system of the set** the wind model follows the wind system that a Hauptwerk set describes: each bellows of the set becomes a wind group of its own, the air consumption per pipe comes from the set, and the middle position of bellows size and maximum drop is that bellows' value. So with a full division, for example, the pedal bellows sags on its own. The controls of the wind model keep working and scale those values; the wind model itself must be on. With **Tremulant follows the set's waveform** the tremulant of a division takes the recorded waveform of a Hauptwerk set as its shape (a measured cycle of pitch and strength) and the set's speed, and each pipe gets the depth the set specifies for it. The speed and depth controls of the tremulant do not apply then; stops with real tremulant recordings do not change.

On the **Loading** tab, the block **Do not load these stops** leaves out stops you do not use when loading, per organ: this saves memory and load time. Open the list, clear the check box of the stops you do not need (each stop shows the memory it needs up front) and choose **Apply and reload**. **Load everything** restores all check boxes. The choice is saved with the organ; an organ without a choice loads everything.

Crescendo

The **general crescendo** draws more stops step by step as you move a crescendo pedal. You decide yourself which stops go in which step:

- **Enabled** switches the crescendo on; **Steps** is the number of steps.
- **Auto-fill** fills in a proposal that suits the stops of this organ. With the options below you decide whether couplers take part, whether colour stops (sesquialtera, cornet, tierce), 32′ reeds and chamades take part and whether the octave couplers only come in at the tutti. **Clear** empties the matrix.
- In the matrix, a click on a cell switches that stop on or off in that step; `Shift` +click sets it for all steps from that step on.
- **Learn pedal** links your crescendo pedal; at the bottom you can also enter channel and CC number by hand.

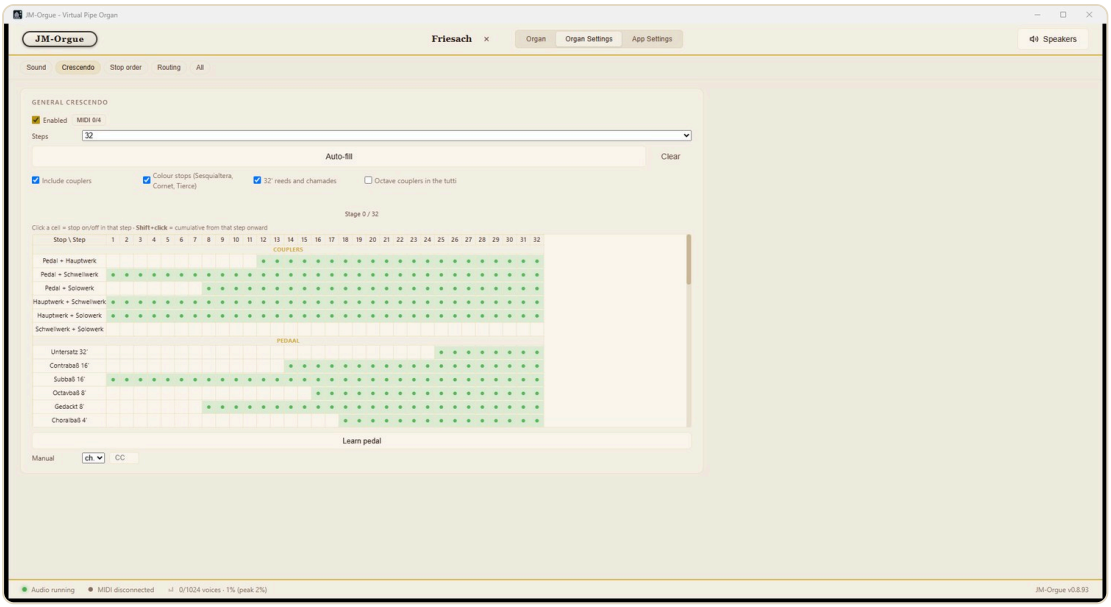


Figure 20. The crescendo matrix: stops in the rows, steps in the columns.

If the set provides a crescendo itself, the button **From the set** appears next to **Auto-fill**: the steps of the set, limited to the stops this organ has and spread over the chosen number of steps. Nothing changes until you press the button, and **Undo** restores the steps.

Stop order

Here you decide the order of the stops in each division. Drag a stop by its handle to move it. For each stop you see the pitch and the family (the colour of the name). With the arrows you change the order step by step, and **Original** restores the order from the organ file. If the automatic classification colours a single stop differently from what you want, you choose the **colour group** per stop yourself: **Foundation**, **Mutation** or **Reed**. That is purely visual and is stored per organ.



Figure 21. The stops per division, in the order in which they appear on the organ screen.

Routing

This tab connects your organ to your equipment. At the top there is a block per keyboard:

- **Channel**: the MIDI channel the keyboard listens to ("All" = every channel). **Learn** determines channel and range by playing the lowest and then the highest key; if those are too close together, the program warns you instead of setting a range of just a few keys. If all your keyboards send on the same channel, give each device a fixed channel under App Settings → MIDI (chapter 8).
- **Transp**: transpose in semitones. **Range**: lowest and highest key. **Short octave**: for sets with a short bass octave.
- **Swell box**: switches on the swell box of this keyboard; buttons then appear to learn the pedal, to set the minimum volume and the cut-off frequency of a closed box. The defaults are −20 dB and 1500 Hz: the pedal then works evenly in decibels over its whole travel, as in Hauptwerk, and the sound becomes somewhat duller with the box closed. If the pedal seems to do little at the bottom, choose a higher cut-off frequency; if you want a closed box softer, a lower minimum volume (down to −24 dB). The pedal can also be learned straight from the organ screen: right-click the green LED strip next to the manual name — or hold it for 2 seconds on a touch screen — and the same "pedal closed / pedal open" dialog opens.
- **Wind supply**: which wind group the keyboard belongs to.
- **Pan** and **Outputs**: the place in the stereo image and the output channels of your audio interface the keyboard sounds on. Two consecutive outputs form a stereo pair; a single channel sounds mono.
- **C/Cis**: spreads the pipes over two sides, like the C and C-sharp chests of a real organ; the global strength and fall-off are in the block above it.
- **Tremulant LFO**: if the set has no real tremulant recordings, the program calculates the tremulant. You set speed, amplitude and pitch here.

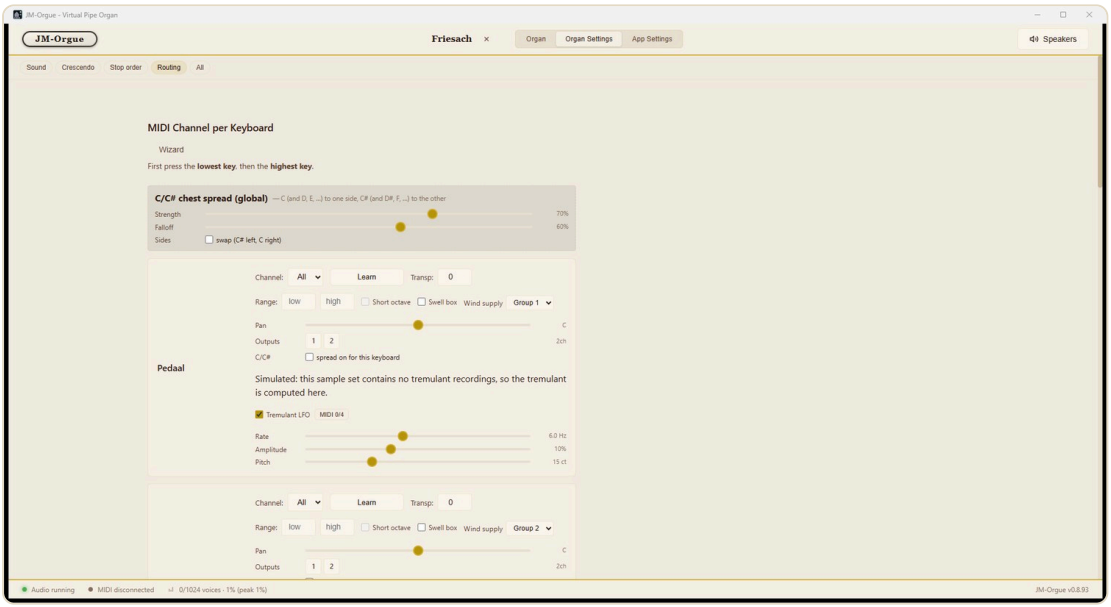


Figure 22. Routing, at the top: the Wizard button, the C/Cis spread and the first keyboard with channel, range, pan, outputs and tremulant.

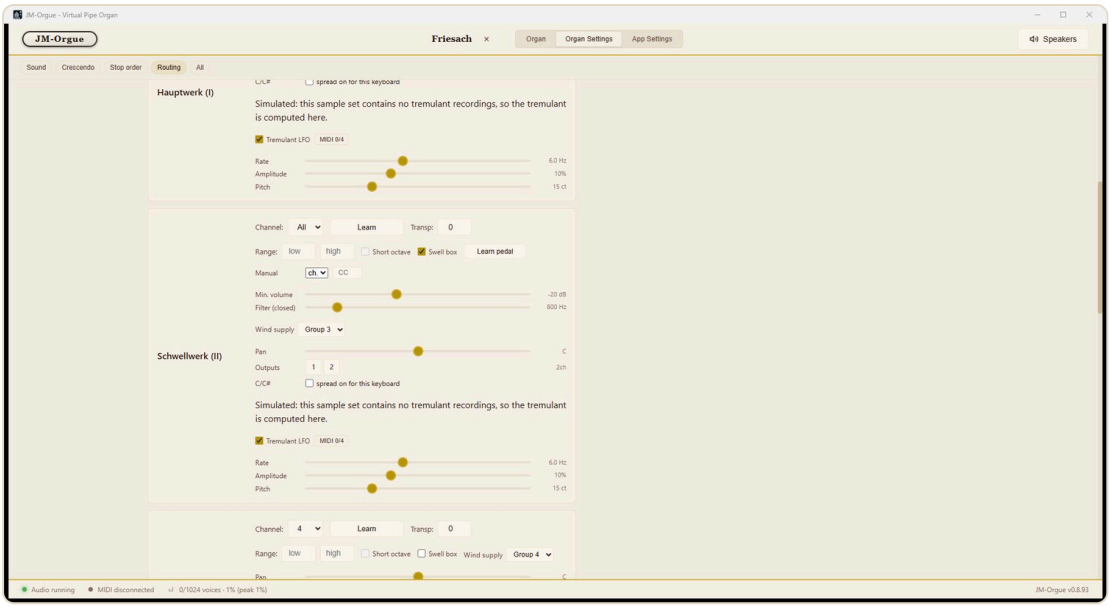


Figure 23. A keyboard with the swell box on: learn pedal, manual channel and CC, minimum volume and cut-off frequency.

Below that is **Background sounds audio output**: for each kind of sound you choose which outputs it uses. **Blower** is the blower sound; **Ambient noise and action clicks** applies to the room noise and the clicks of stops and keys; sets with effect stops (cymbelstern, nightingale and the like) and tremulant noise get a row of their own. This matters mostly with a set-up of more than two outputs, where the keyboards sound on other channels than the default 1–2. Directly below:

- **Continuous keyboard**: keys outside the recorded range of a stop play the nearest pipe, shifted to the right pitch, instead of staying silent. A 61-note keyboard thus keeps playing on a set with fewer notes. Bass and treble halves of a divided stop (Trompet 8 B/D, Cornet D, Mixtuur Bas …) stop at the split point — the treble carries on at the top, the bass at the bottom, but they never overlap.
- **Automatically retune pipes**: retunes the pipes based on their measured pitch, while keeping deliberately floating stops. On by default.
- **Pitch correction**: everything together higher or lower in cents (**Zero** resets it).

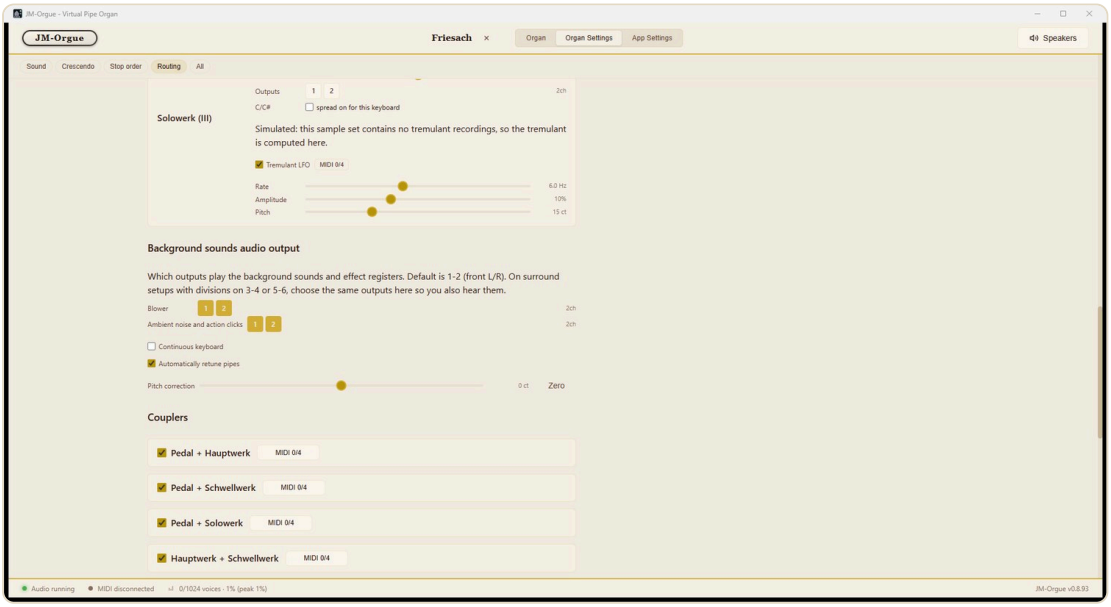


Figure 24. Background sounds audio output with the chips per output channel, continuous keyboard and pitch correction.

Further down are **Couplers** (switch on or off, with a MIDI button per coupler) and **Extra couplers**: couplers that are not in the sample set but follow from the keyboard layout; tick what you want and they appear straight away on the coupler bar. **MIDI bindings** is one list of all MIDI bindings of this organ, which you can adjust, remove or add by hand there. At the bottom, **Open .mid** lets the organ play a MIDI file.

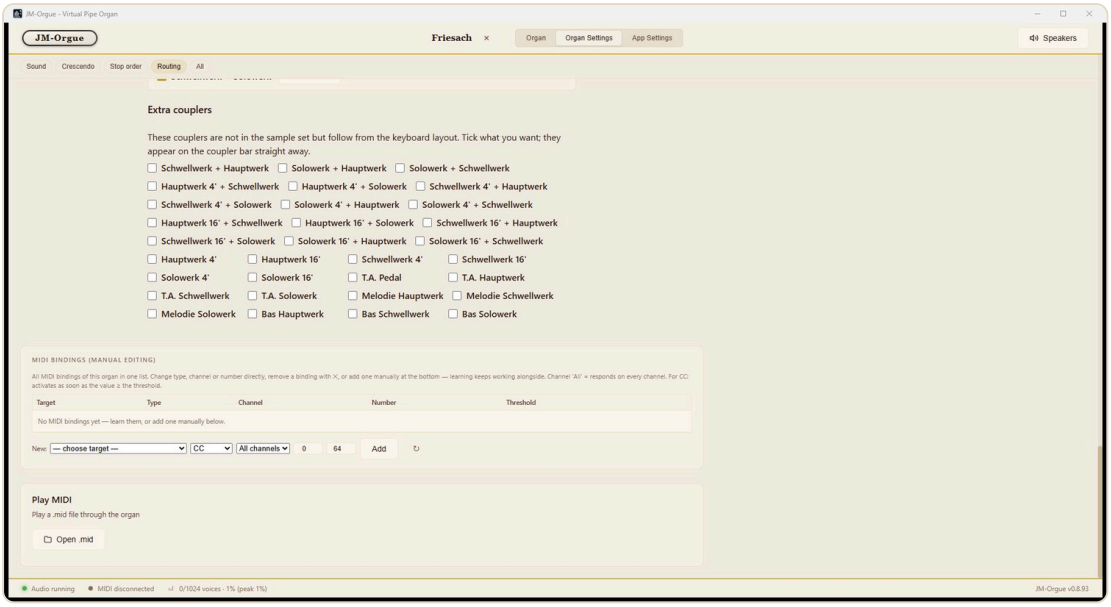


Figure 25. Routing, at the bottom: extra couplers, MIDI bindings and Play MIDI.

8. App Settings

These settings apply to the whole program. The tabs are **Audio output**, **Sample fix**, **MIDI**, **Learn buttons**, **Display** and **Update and feedback**.

Audio output

- **Audio driver** and device: for the lowest latency choose ASIO (for example the driver of your audio interface); WASAPI works with the standard Windows sound card. The “Current” line below shows what is playing now.
- **Test signal**: click a channel number to send sound to exactly that output, so you hear which plug belongs to which loudspeaker.
- **Polyphony** and **Calculation cores for mixing**: the maximum number of sounding pipes and the number of processor cores the mixing work is spread over. The default choice fits your computer; the load meter shows the effect.
- **Play stereo samples in stereo**: keeps the stereo image of stereo recordings. Off means a mono mix and halves the memory use of such sets.
- **Start buffer per recording**: how much of each pipe recording is in memory straight away: 200 ms, 300 ms, 500 ms, 1 s or 2 s (default). A shorter buffer saves a lot of memory and loading time; the rest is loaded while you play. The choice takes effect after the organ is reloaded. Once enough recordings have loaded, a **suggestion** for this PC appears below it, based on the measured loading times; **Apply** adopts it. Nothing changes by itself.
- **Buffer size**: smaller means less latency but is harder on your computer. **Apply** activates the choice.

> **TIP** If you hear a short repeat of the attack of a pipe in big chords, choose a longer start buffer. If you do not hear that but have little memory, 500 ms or 1 s is a good choice. A fast drive (SSD) makes short buffers possible.

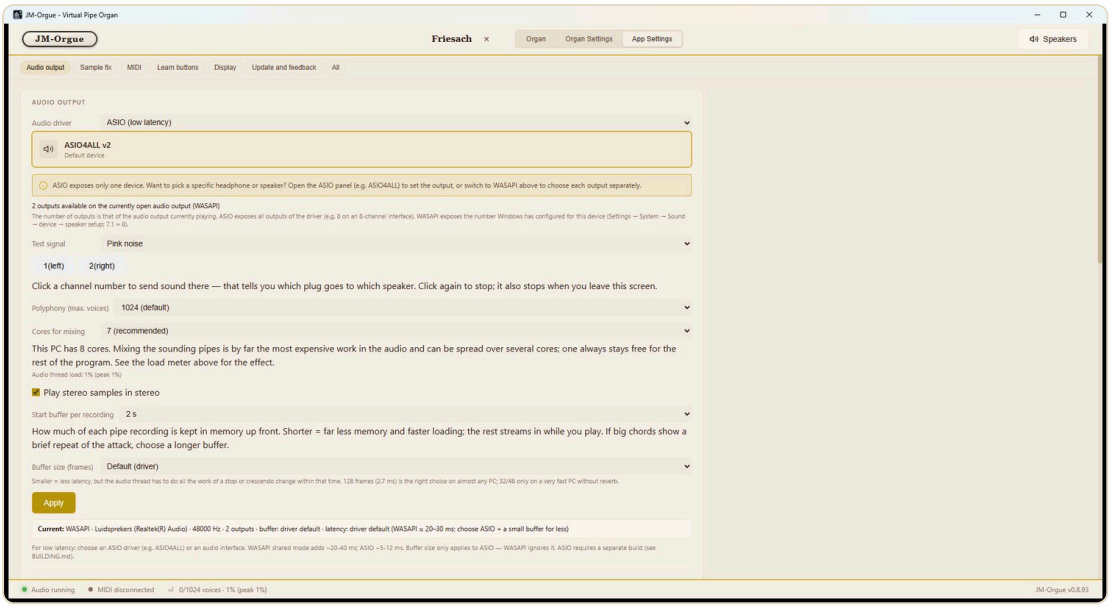


Figure 26. Audio output, at the top: driver, device, test signal, polyphony, calculation cores, stereo samples, start buffer and buffer size.

More under Audio Output

- **Low frequencies to a subwoofer (bass management)**: takes the low frequencies out of the main channels and sends them, summed, to one subwoofer output (a fourth-order Linkwitz-Riley filter). You choose the **Subwoofer output**, the **Crossover frequency (Hz)** (30 to 250 Hz) and the **Subwoofer level (dB)**. Off by default; off costs nothing.
- **Memory limit follows free memory**: the ceiling for fully loaded samples follows the memory that is free when an organ is loaded.
- **Real-time priority for mixing**: gives the mixing workers a high priority so the audio does not stutter when much else is running. Below the check box it shows what is active.
- Behind each choice of **Start buffer per recording** is the memory the loaded organ then needs up front, for example “500 ms · 2.8 GB”.
- **Samples in memory**: **16-bit compact (default)** also keeps samples that fit exactly in 16 bits as 16 bits, with no difference in sound: half the memory for those sets. **16- and 24-bit compact** also saves a quarter on 24-bit sets, but costs noticeably more processing; **32-bit (as before)** turns it off. Takes effect after the organ is reloaded.
- **Disk cache of decoded samples** (on by default): recordings that are expensive to produce, such as WavPack or recordings converted to the sample rate of the output, are kept on disk after the first time and then load much faster. They sound exactly the same. With **Maximum** you set the size, below it is shown how much is in use, and **Clear cache** empties it.

Output profiles store two complete output settings, for example loudspeakers via ASIO and headphones via WASAPI, including the channel choice per keyboard. Choose the output you want, click **Save current selection** at Speakers or Headphones, and then switch with the **Speakers** button at the top right or with a learned MIDI button. The profile that is active now is marked.

Correction for headphones or speakers applies a tone correction to the output itself (not to the organ): choose which profile it applies to, switch it on and choose a preset from the AutoEq list, or import or export a correction of your own.

Below that are the options to start the program automatically with Windows, to load the last opened organ automatically, to let the cross on an extra screen close the whole software and to restore the last registration. The **Automatic MIDI archive** records in the background everything you play, as a .mid file per session: it starts at the first note and stops after a silence.

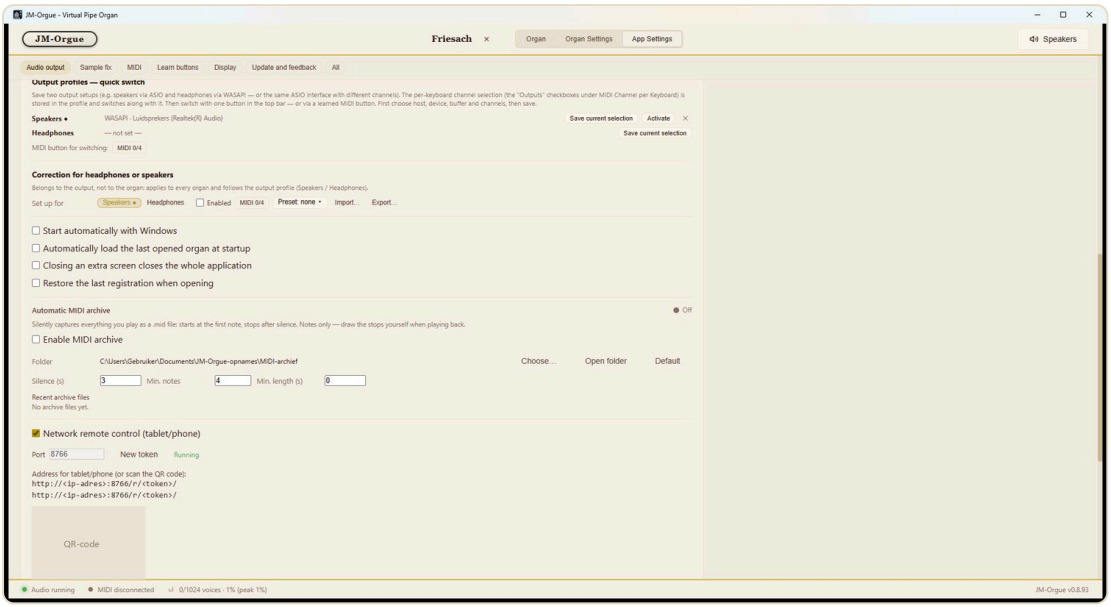


Figure 27. Audio output, further down: output profiles, correction for headphones or speakers, start-up options, MIDI archive and remote control. The folder and the network address in this image have deliberately been replaced by a placeholder.

With **Network remote control (tablet/phone)** you control the organ from a tablet or phone on the same wi-fi network. The program shows an address and a QR code to scan, and you choose which parts and which divisions appear on the external screen. The first time, Windows Firewall asks whether JM-Orgue may receive connections: choose “Allow access” for private networks.

> **NOTE** The address contains an access token and works only within your own network, without encryption. Do not share the address or QR code; with **New token** you withdraw the access of a device that scanned it earlier.

At the bottom is the **sample rate**. It follows the sound card itself: under ASIO you set it in the driver's panel, under WASAPI in the Windows sound settings.

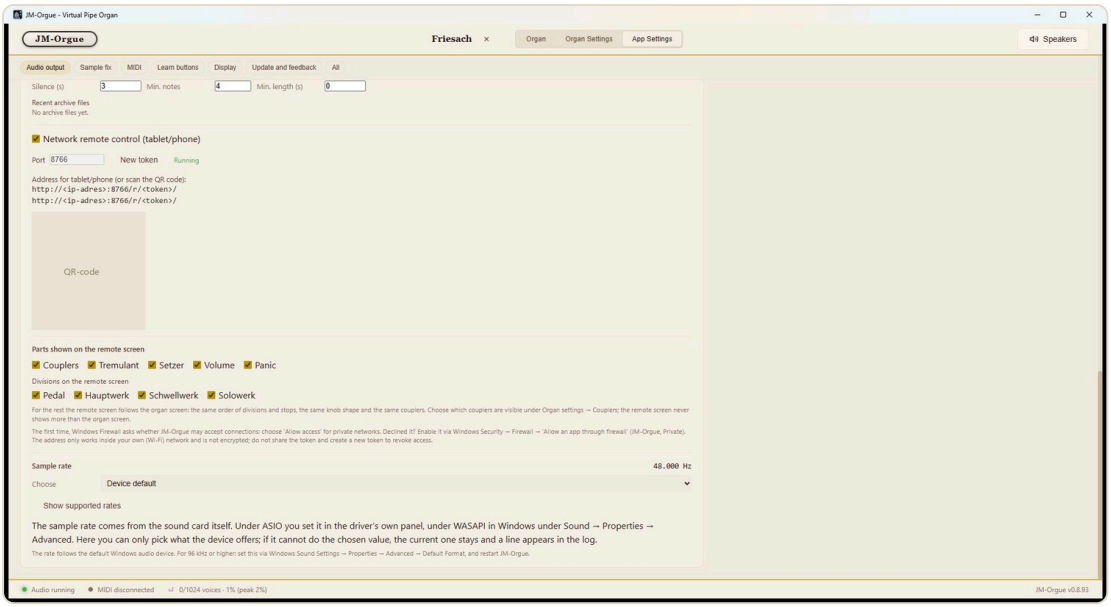


Figure 28. Audio output, at the bottom: the remote control with parts and divisions on the external screen, and the sample rate. The QR code and the address are covered here.

Sample fix

Two tools to tidy up the samples of the loaded organ. **Leading silence** finds and cuts silence at the start of the samples (you set the silence threshold in dB and the lead-in you want to keep yourself); **Loops** finds a good sustain loop, bakes in a crossfade for a seamless transition and writes the loop points into the file, with the option to do this only for files without a loop. With **Scan** you first look at what would change. The originals are first kept in a folder _backup before the files are overwritten.

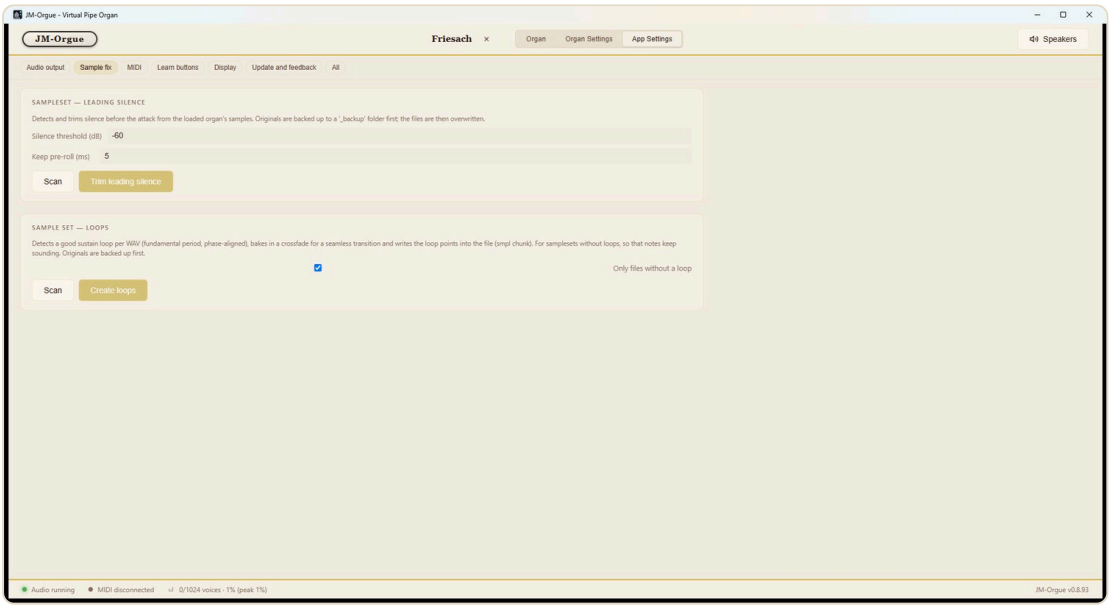


Figure 29. Sample fix: leading silence and loops, each with Scan and a button to apply.

MIDI

MIDI Input shows the connected MIDI devices (with **Refresh devices** you search again). Per device you can choose a fixed channel: keyboards that all send on channel 1 then still operate different divisions, without you having to restore channels every session. If a device is set to **Default device**, nothing changes. Under **Bluetooth MIDI** you search for wireless devices with **Scan**; the learn buttons that react

to a single key also listen to a Bluetooth keyboard. **Feedback (MIDI out)** makes your console light up with the registration: choose the protocol (stop lamps via Note On/Off, a Hauptwerk text display or a Johannus/Content console), the MIDI output, the channel and the base note. Each stop then automatically gets a note number from the base note, or you learn the real console button per stop with **Learn per stop…**.

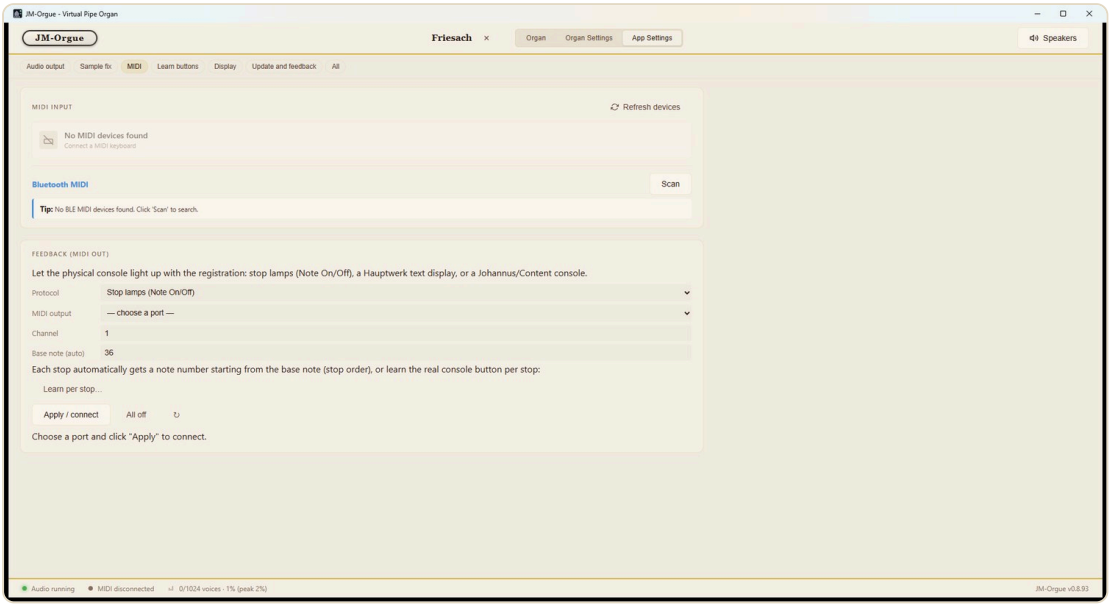


Figure 30. MIDI input, Bluetooth MIDI and the feedback to the console.

Learn buttons

Here you learn the fixed console buttons (pistons): tuner, SET, preset steps, memory levels and shutting down the computer. The digit buttons 0–9 you learn on the setter bar itself (right-click or hold for 3 seconds).

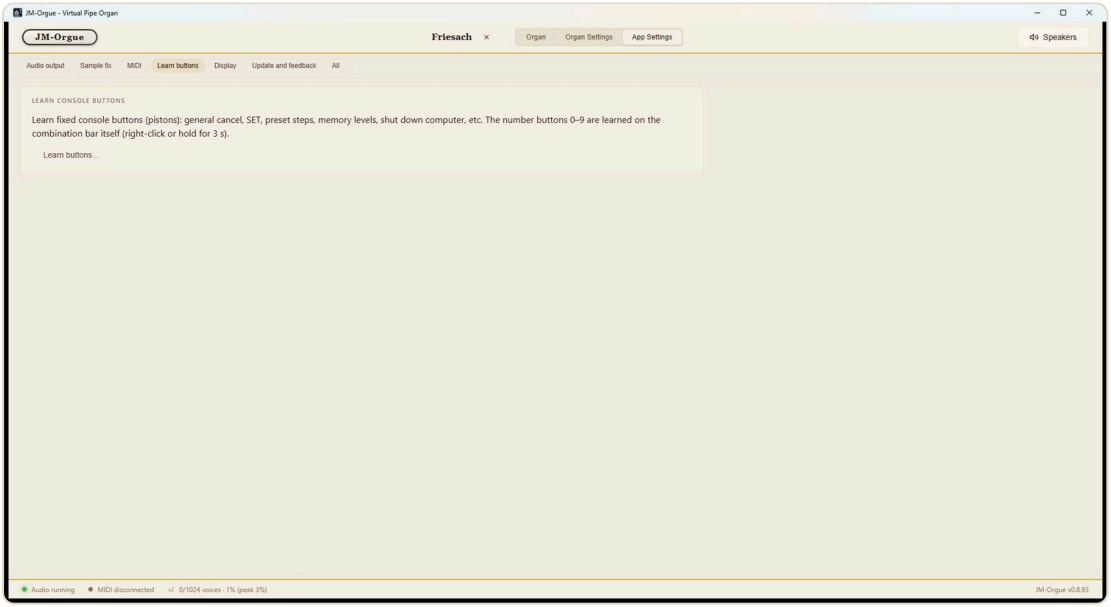


Figure 31. Learning console buttons.

Display

Choose the language of the program (Nederlands, English, Français, Deutsch, Polski, Italiano and Español) and the look: five ready-made colour presets (Basic, Classic wood, Modern minimal, Night and

Evening), or put together your own colours, give them a name and export them. Below that you adjust the font, size and weight of the stop text, and the background.

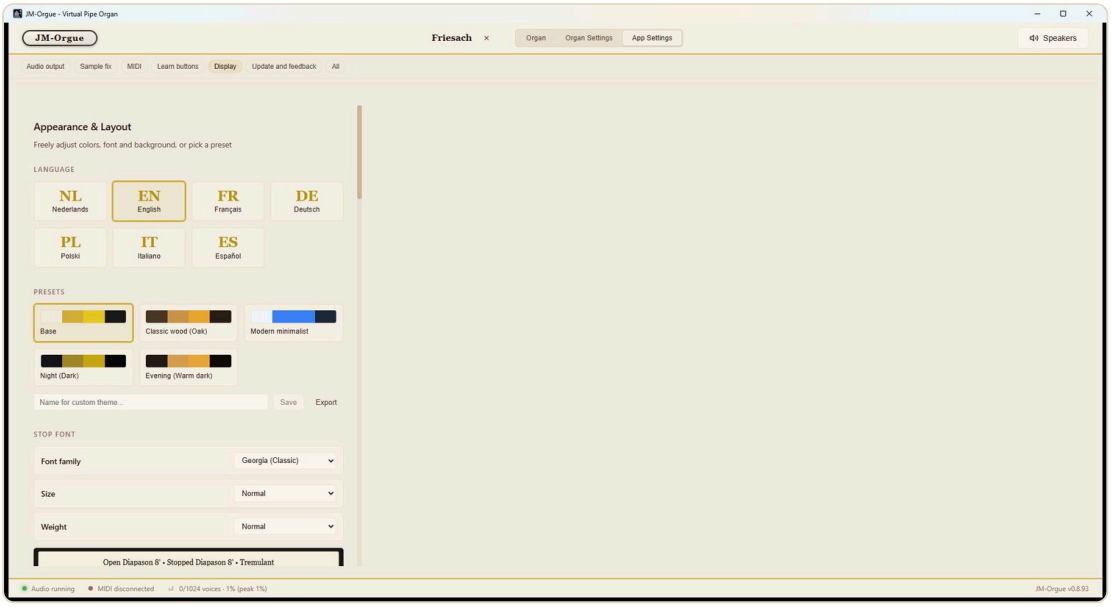


Figure 32. Appearance & Layout: language, presets and font for the stops.

Update and feedback

Here you find the version number, the credits for the headphone presets and loudspeaker data used, and the buttons **Send feedback…**, **Via GitHub (issues)** and **Check for updates**.

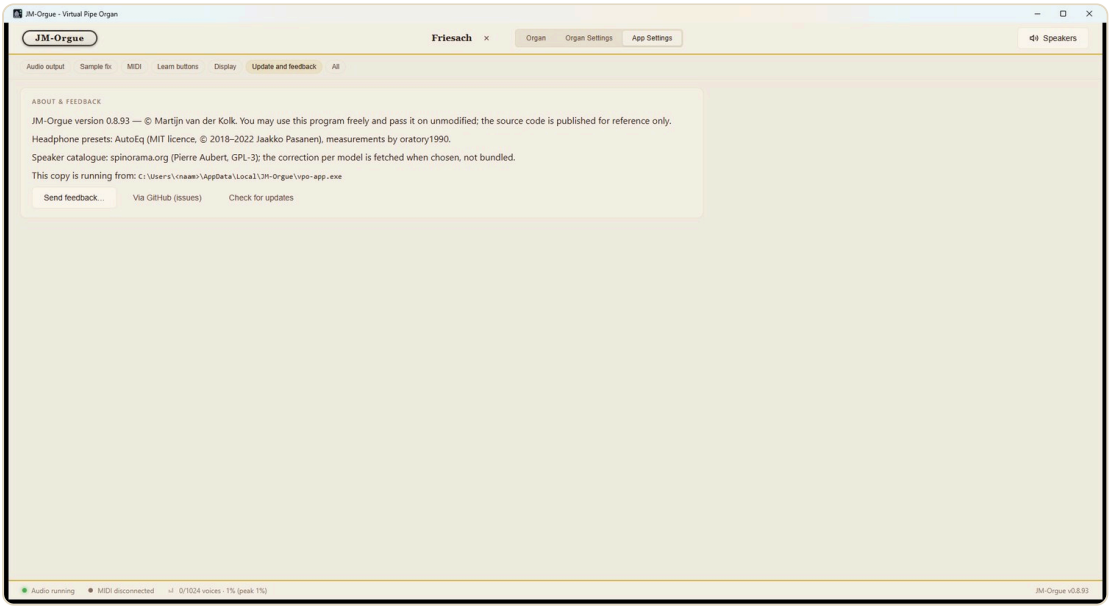


Figure 33. About & feedback.

9. Acoustic configuration of a set

An organ definition can refer to an extra file, `JM-ACOUSTICS.json`, that describes the acoustics of the set: swell boxes attached to a rank instead of to a keyboard, impulse responses for the reverb, mixing routes and outputs. If your set has such a file, you recognise it by an extra block in Organ Settings → Sound.

- The program reads the file and checks it: the file size, whether all references are correct, whether the paths stay inside the organ folder and whether the pipe bindings fit the loaded definition. If something is wrong, the organ simply plays with the regular definition and the block says why the configuration was not loaded.
- **Swell boxes per rank**: ranks that the file puts in a box get a swell box and swell pedal of their own, independent of the keyboard they are played on. A rank used by two divisions thus always follows the same box. You learn the pedal under Routing (**Learn pedal**) or in the wizard.
- **Impulse responses** from the set are listed under the block, with channels and duration. With **Use as reverb** you adopt one as convolution reverb. The role of the channels in such recordings is not defined; the program uses the first two as a stereo pair.
- Mixing routes, output buses and effect transitions from the file are for now not applied automatically; the regular definition remains in charge.

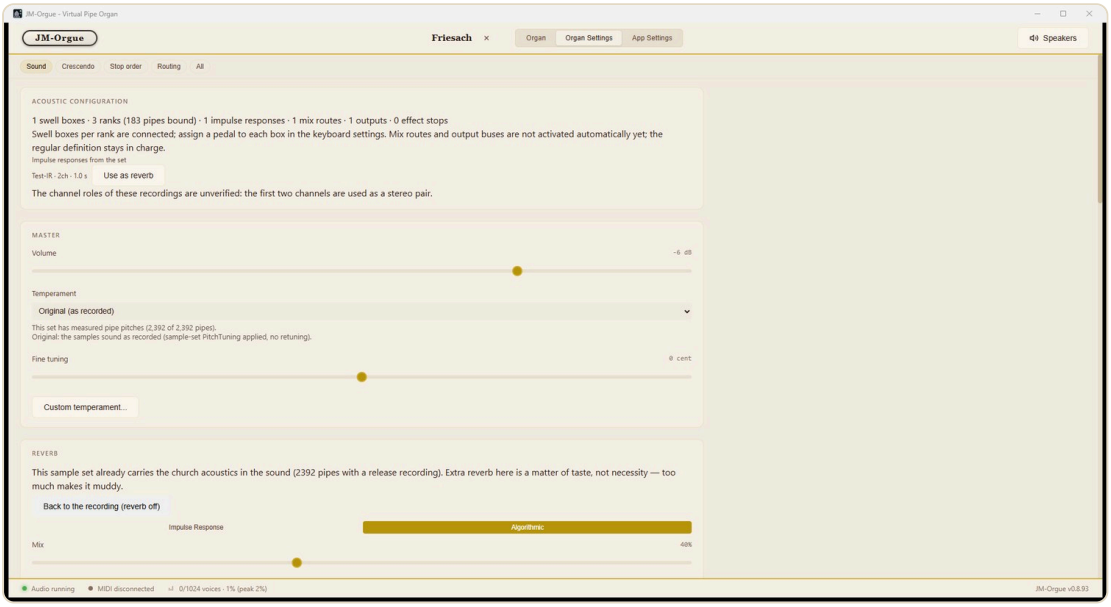


Figure 34. The “Acoustic configuration” block at the top of Sound, with a summary and a button per impulse response. The images in this chapter were made with a temporary test file for a demo organ.

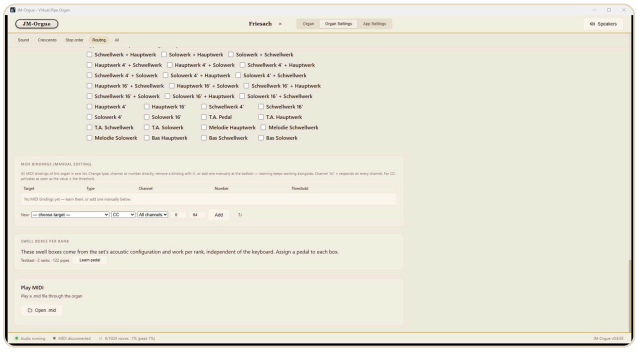


Figure 35. Routing, at the bottom: the Swell boxes per rank block with a “Learn pedal” button per box.

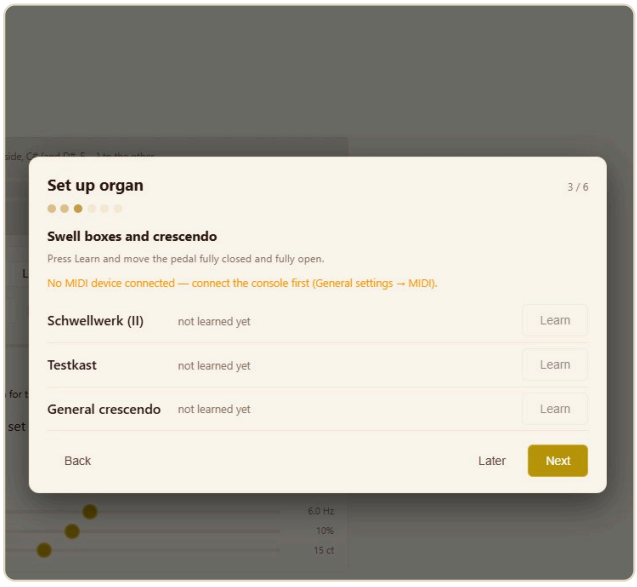


Figure 36. The wizard shows the box under the swell boxes and the crescendo.

10. Keyboard shortcuts and tips

Key	Function
`F1` `F2` `F3`	Organ, Organ Settings and App Settings.
`F11`	Full screen (console mode) on or off.
`Esc`	Full screen off.
Right-click or hold 3 s	On a button of the setter bar: link that button to a console button (MIDI learn).
Double-click on the volume	Master volume back to the default value.
Card held for 3 s	In the library: search the photo of that card again.
`Z` … `M` and `Q` … `I`	Computer keyboard as a manual: two octaves. `Shift` shifts an octave up, `Alt` an octave down, also combined.
`Ctrl` + `N` / `O` / `S`	In the notation window: new piece, open, save.

- All settings are stored with each organ, also when you close the organ or quit the program.
- For a fixed console: switch on **Start automatically with Windows** and **Load last opened organ automatically**, and use console mode.
- With a set-up of more than two outputs: choose the outputs per keyboard under Routing, and then the same outputs for the background sounds.

11. Troubleshooting and feedback

Problem	What you can try
No sound	Is the blower on? Does it say “Audio active” at the bottom? Is the right device set under Audio output, and do the outputs per keyboard match your audio interface? Use the test signal to hear the outputs.
The keyboard does not react	At the bottom it says “MIDI not connected” as long as there is no MIDI device. Check App Settings → MIDI and the MIDI channel per keyboard under Routing.
The organ sounds out of tune	Check that **Automatically retune pipes** is on, and under Sound choose the temperament “Original” or the temperament you want.
Crackling or stuttering	Lower the polyphony, draw fewer stops, switch the reverb off or choose a larger buffer. A short start buffer needs a fast drive; otherwise choose a longer one.
Too little memory	Choose a shorter start buffer, switch stereo samples off, or switch off listening positions you do not use.
Background sounds or effect stops are not heard on the right speaker	Under Routing, at “Background sounds audio output”, choose the same outputs as for the keyboards.
Pipes stay silent and a notice about WavPack or FLAC appears at the top	WavPack the program reads by itself; if the notice persists, it concerns files that cannot be read (FLAC, hybrid or multichannel WavPack). Convert the set to plain WAV or ask the maker for an unpacked version.
The swell pedal does little at the bottom and a lot at the top	Under Routing, at the swell box, raise the cut-off frequency for a closed box (default 1500 Hz) or lower the minimum volume.
An organ loads by accident or takes too long	Press **Cancel** in the loading screen; the previous organ keeps sounding.

Send feedback… (App Settings → Update and feedback) creates a message with your report and the log files of the last sessions, so the developer can see what happened. You can also post your report on the project page on GitHub.